

MUHAMMAD SONY SASTRANTARA

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Detail oriented Master's graduate and GIS Specialist with robust expertise in remote sensing, marine ecosystem mapping, and geospatial analysis. Adept at leveraging drone technology and satellite imagery for comprehensive spatial mapping and data interpretation to drive environmental conservation initiatives. Proven track record in executing complex field surveys, photogrammetry, and habitat analysis. Possesses a strong foundation in marine resource management, eager to leverage advanced mapping technologies and data-driven insights to support impactful, sustainable environmental projects.

WORK EXPERIENCE

OceanKita

Oct 2024 - Now

GIS Expert

- **Survey Operations:** Orchestrated and executed comprehensive drone-based field surveys across 13 river mouths in Jakarta, optimizing spatial data collection efficiency.
- **Project Management:** Spearheaded environmental initiatives at designated river sites, overseeing project planning, execution, and continuous monitoring to ensure operational success.
- **Stakeholder Engagement:** Engaged and collaborated with coastal communities to align strategies and execute targeted field efforts for marine debris management.
- **Data Analysis:** Analyzed high-resolution drone imagery to detect and quantify macroplastic pollutants and organic waste accumulation in river ecosystems.
- **Reporting & Insights:** Processed complex geospatial data and synthesized survey findings into comprehensive reports, delivering actionable insights and recommendations for environmental management.
- **Global Collaboration:** Partnered with international stakeholders, including Global MeteOcean and CQUniversity Australia, to present field data, evaluate findings, and drive strategic environmental solutions.

PT. Berkat Ecolestari Sejahtera

Oct 2023 – Des 2023

Freelance GIS Technician

- **Spatial Mapping:** Conducted precise polygon mapping of small Indonesian islands utilizing the latest satellite imagery for a national island inventory project.
- **Data Classification:** Analyzed structural formations, area dimensions, and island typology to support accurate geospatial database development.

ITK IPB

Jan 2023 – Des 2023

Assistant

- **Technical Instruction:** Facilitated marine-focused practical sessions, instructing students in remote sensing techniques utilizing visible light sensors for coastal and marine ecosystem mapping.
- **Thermal Analysis:** Mentored students in the acquisition and analysis of thermal satellite data for sea surface temperature mapping and the detection of critical thermal phenomena.
- **Image Processing:** Delivered instruction on digital image processing, including noise reduction and image enhancement, applied to benthic habitat identification and water quality monitoring.

- Field Survey Support: Supported the design and execution of marine resource surveys, demonstrating the application of GIS and drone technology for accurate spatial data collection and sustainable mapping.

EDUCATION

Master of Science Aug 2023 - Sep 2024
Bogor Agricultural Institute (IPB)

- Major in Marine Technology (GPA 3.93 of 4.00)
- Specialize in object detection using remote sensing tools
- Fastrack Program

Bachelor of Marine Science and Technology Aug 2019 - Aug 2023
Bogor Agricultural Institute (IPB)

- Major in Marine Science and Technology (GPA 3.37 of 4.00)
- Specialize in marine remote sensing and GIS

Publications

Sastraantara, M. S., Agus, S. B., & Susilo, S. B. (2024). Mapping the distribution of sea urchin (Echinoidea) and benthic habitat using drone in the waters of Lancang Island. BIO Web of Conferences, 106, 04006.

ADDITIONAL INFORMATION

Languages : Bahasa Indonesia, English (TOEFL ITP Score of 587)

Certifications : Remote Pilot Competencies Certification by APDI (2024-2026), Remote Pilot Certification by The Directorate General of Civil Aviation Indonesia (2025-2026)

Technical Skills: ArcGIS, QGIS, Word, Excel, PowerPoint, Photogrammetry, Drone Survei, Drone Imaging, Drone Mapping, GIS Software, Remote Sensing Software, Image Analysis Software, Computer Hardware.

Hardware Proficiency & Owned Equipment: DJI Mavic Pro, Data Processing Workstation